contested and surveillance-sensitive environments, forming a critical foundation for future European defense, civil aviation security, and humanitarian drone operations.

9.8.12 Verizon Skyward – 5G drone operations in the United States

Verizon’s Skyward platform played a pioneering role in demonstrating how 5G cellular networks can support real-time, long-range, and data-intensive drone operations in the United States. Before its program concluded in 2022, Skyward worked in partnership with commercial operators, industrial clients, and public safety agencies to prove that drones connected through carrier-grade cellular networks could maintain stable command-and-control links, stream high-resolution imagery, and coordinate multi-drone missions beyond visual line of sight (BVLOS). These trials marked a significant step toward integrating drones into national airspace and developing scalable urban and regional drone service networks.

Skyward’s core contribution was the operational validation of network-based flight control and telemetry transmission. By leveraging Verizon’s 5G infrastructure, drones were able to access low-latency, high-bandwidth uplink and downlink channels, allowing operators to receive continuous situational data, detect hazards, and adjust mission parameters dynamically. This capability proved especially valuable in industrial asset inspection, emergency response, utility monitoring, and real-time event surveillance, where uninterrupted data flow is essential for decision-making and safety.

The program also contributed to the development of unmanned aircraft system traffic management (UTM) integration frameworks in the United States, working alongside the Federal Aviation Administration (FAA) and NASA. Skyward’s testing helped shape protocols for flight approvals, dynamic airspace awareness, and automated remote pilot notifications,